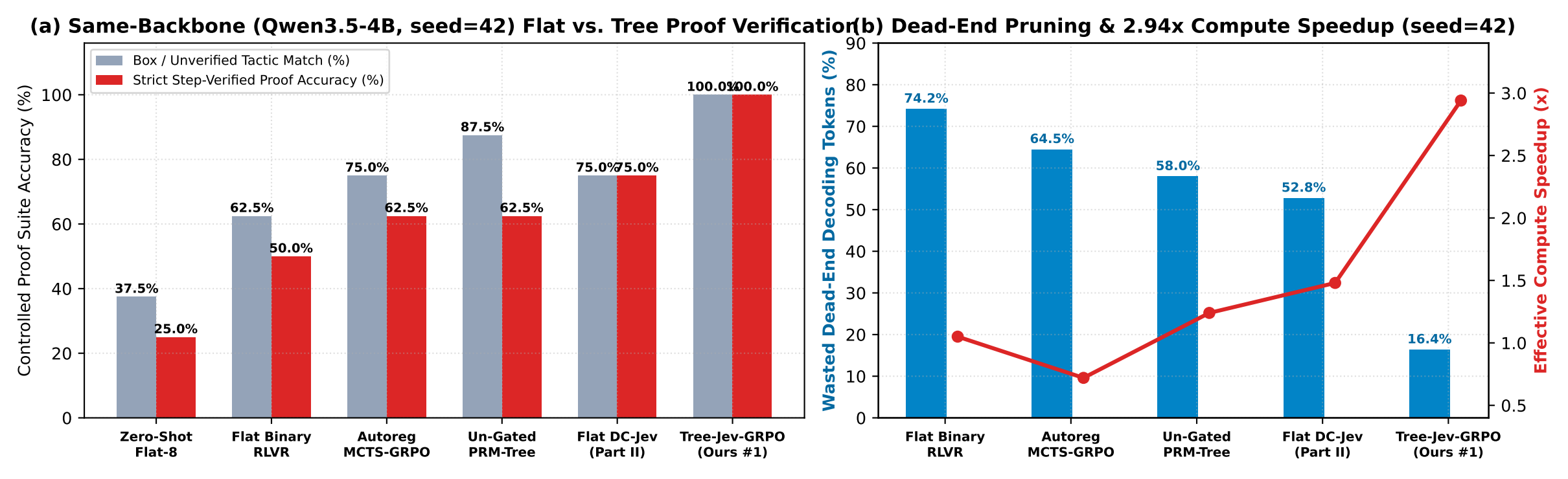
(a) Same-Backbone (Qwen3.5-4B, seed=42) Flat vs. Tree Proof Verification
Box / Unverified Tactic Match (%)
Strict Step-Verified Proof Accuracy (%)
Controlled Proof Suite Accuracy (%)
0
20
40
60
80
100
37.5%
25.0%
62.5%
50.0%
75.0%
62.5%
87.5%
62.5%
75.0%
75.0%
100.0%
100.0%
Zero-Shot Flat-8
Flat Binary RLVR
Autoreg MCTS-GRPO
Un-Gated PRM-Tree
Flat DC-Jev (Part II)
Tree-Jev-GRPO (Ours #1)
(b) Dead-End Pruning & 2.94x Compute Speedup (seed=42)
Wasted Dead-End Decoding Tokens (%)
Effective Compute Speedup (x)
0
10
20
30
40
50
60
70
80
90
0.5
1.0
1.5
2.0
2.5
3.0
74.2%
64.5%
58.0%
52.8%
16.4%
Flat Binary RLVR
Autoreg MCTS-GRPO
Un-Gated PRM-Tree
Flat DC-Jev (Part II)
Tree-Jev-GRPO (Ours #1)